Lab 3

A Look Under the Hood: the Command Line

Day 1 · 15:45–16:45 · 60 min
Pairs · swap driver and navigator at Part C

LO3: understand what Galaxy does behind the buttons, and take a first step towards independent analysis

Open your handout

Lab 3 handout

`Lab3_command_line_taster.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

Why bother?

- Every Galaxy tool you clicked today is a **command-line program**
- A dozen commands let you look inside huge files and follow online tutorials

KEY MESSAGE

You don’t need to become a programmer. This is the most durable skill in the course.

Golden rules

- Commands are **case-sensitive**, and spaces matter
- **Enter** runs a command · ↑ repeats the last one
- **Tab** completes file names
- **Ctrl + C** stops a running command

Open a terminal inside Galaxy

1. Run the tool **Interactive JupyterLab Notebook** (no changes)
2. After 1–3 min: **User** → **Active InteractiveTools** → click the link
3. In JupyterLab: **File** → **New** → **Terminal**
4. You'll see a prompt ending in `$`. Type here.

> Not working? Use your laptop's **Terminal** (macOS) or **Git Bash** (Windows).

Where am I? Moving around

Command	What it does
`pwd`	**p**rint **w**orking **d**irectory: where am I?
`ls`	list the files here
`mkdir ssi_course`	make a new folder
`cd ssi_course`	go into a folder
`cd ..`	go up one level

PART B

Try it

Type each line, then press Enter:

```
pwd
ls
mkdir ssi_course
cd ssi_course
cd ..
pwd
cd ssi_course
```

CHECK YOURSELF

Draw a little “tree” of the folders you moved through.

Swap driver and navigator

The navigator takes the keyboard; the driver reads the handout

Download the reads

The same Illumina read 1 file you used in Lab 2:

```
wget https://zenodo.org/…/DRR187559_1.fastqsanger.bz2
ls -lh
```

- Copy the **full URL** from your handout (Part C1)
- `ls -lh` lists files with their sizes (K, M, G)

PART C2

Peek inside a compressed file

```
bzcat DRR187559_1.fastqsanger.bz2 | head -8
```

- `bzcat` unzips the file as it reads it
- `head -8` shows the first 8 lines
- `|` is a **pipe**: it sends one command's output into the next

CHECK YOURSELF

How many reads did you see? Point to the first read's quality line.

PART C3

Count the reads

FASTQ has 4 lines per read: **count lines, then divide by 4**

```
bzcat DRR187559_1.fastqsanger.bz2 | wc -l
```

CHECK YOURSELF

How many reads? Does it match “Total sequences” in your Lab 2 FastQC report?

PART C3

A trap

Someone suggests counting reads like this:

```
bzcat DRR187559_1.fastqsanger.bz2 | grep -c "^@"
```

It counts lines that start with `@`.

> **CHECK YOURSELF**
>
> Why could this give the **wrong** answer?
> *Hint: which quality score is @?*

PART C4

Search inside the file

How many reads contain a run of 10 A's?

```
bzcat DRR187559_1.fastqsanger.bz2 | grep -c "AAAAAAAAAA"
```

`grep` finds lines containing a pattern; `-c` counts them

From reads to genomes

Download the assembly (URL in your handout, Part D), then:

```
head -3 DRR187559_contigs.fasta
grep -c ">" DRR187559_contigs.fasta
```

CHECK YOURSELF

What does `grep -c ">"` count? Compare it with **# contigs** in QUAST.

Galaxy ↔ command line

In Galaxy…	…on the command line
Upload → Paste URL	`wget URL`
Eye icon	`head`, `less`
Line/Word/Character count	`wc -l`
History	`history` (try it!)

> When you finish: **User** → **Active InteractiveTools** → **Stop** JupyterLab.

Cheat sheet 1: moving around

`pwd`	where am I?
`ls` · `ls -lh`	list files · with sizes
`cd folder` · `cd ..`	go into a folder · go up
`mkdir name`	make a folder
`wget URL`	download a file
`history`	commands you have run

Cheat sheet 2: looking inside files

`head -n 8 file`	first 8 lines
`wc -l file`	count lines
`grep "text" file`	find lines containing text (`-c` counts them)
`bzcat · zcat`	read a .bz2 · .gz file without unzipping
`cmd1 | cmd2`	pipe: send cmd1's output into cmd2

Want more? Software Carpentry **Unix Shell** lesson: `swcarpentry.github.io/shell-novice`

Adapted from The Carpentries "The Unix Shell" and the GTN "CLI basics" tutorial (CC-BY 4.0).